

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Case Study

COURSE NAME:	Cloud Computing and Big Data Analytics	COURSE CODE:	CSA15
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COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Dave's Taxonomy: Precision (Level 3)
Question Count: 3

Total Marks: 30

Code of Conduct

I Srinivas S certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Srinivas S

Assessment Tasks

Case Study

1. Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

2. Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

3. Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced

through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Case Study 1: Smart Retail Inventory Management System

Answer:

The Smart Retail Inventory Management System is a cloud-based solution designed to manage product inventory, transactions, and demand forecasting across multiple retail stores.

1. Requirements

- POS systems and dashboards should communicate with centralized cloud services.
- Product data, transaction logs, and analytics datasets must be stored securely.
- The system should support real-time inventory updates.
- Analytics and demand forecasting should be provided through cloud platform services.
- Secure communication should be maintained using VPNs, firewalls, and API gateways.

2. Cloud Architecture

The architecture can be designed as:

Store Devices → API Gateway → Cloud Services → Cloud Storage

- **Clients:** POS systems and management dashboards.
- **Web APIs:** Enable communication between clients and cloud services.
- **Cloud Storage:** Stores product information, transaction records, and analytics datasets.
- **Platform Services:** Perform analytics and demand forecasting.
- **Security:** VPNs, firewalls, and API gateways protect communication and services.

3. Application of Cloud Concepts

Cloud infrastructure allows the retail chain to centralize its data instead of maintaining separate systems at every store. Platform services can process historical sales data and generate demand forecasts.

For example, if a product is frequently sold during a particular period, the analytics service can identify the pattern and help the retailer maintain sufficient stock.

Conclusion:

The proposed cloud-based inventory system provides an efficient and scalable solution for managing retail inventory. The integration of APIs, cloud storage, analytics services, and security mechanisms makes the system suitable for real-world retail operations.

Case Study 2: Cloud-Based Document Collaboration System

Answer:

The Cloud-Based Document Collaboration System is similar to an online document editing platform where multiple users can access, edit, and share documents through web browsers.

1. Requirements

- Users should access documents through browsers.
- Multiple users should be able to collaborate in real time.
- Documents should be stored in distributed cloud storage.
- Version control should preserve previous document versions.
- Web APIs should synchronize changes between devices.
- The system should provide security through encryption and identity management.

2. Cloud Architecture

The architecture can be represented as:

Users/Browsers → Web APIs → Cloud Platform → Distributed Cloud Storage

- **Clients:** Web browsers and other supported devices.
- **Web APIs:** Synchronize document changes between clients.
- **Cloud Storage:** Stores documents in distributed storage.
- **Version Control:** Maintains different versions of documents.
- **Network Infrastructure:** Provides low latency and high availability.
- **Security:** Encryption, identity management, and access permissions protect documents.

The case specifically states that users access the platform through browsers, documents are stored in distributed cloud storage, and APIs provide synchronization across multiple clients and devices.

3. Application of Cloud Concepts

Distributed cloud storage allows users to access their documents from different locations and devices. Web APIs allow modifications made by one user to be synchronized with other users.

Version control is important because users can recover previous versions if an incorrect modification is made.

Conclusion:

The cloud-based document collaboration system provides an efficient platform for real-time document creation, editing, and sharing. Distributed storage, APIs, version control, and security mechanisms make it suitable for enterprise collaboration.

Case Study 3: Online Food Ordering Platform**Answer:**

The Online Food Ordering Platform is a cloud-based application that allows customers to browse menus, place orders, and receive services through web and mobile applications.

1. Requirements

- Customers should access the application through browsers and mobile devices.
- Frontend clients must communicate with backend services.
- RESTful APIs should handle communication.
- Menu images, invoices, and customer information should be stored securely.
- The system should support high numbers of users.
- Security should include authentication and role-based access control.
- The system should be reliable and scalable.

2. Cloud Architecture

The architecture can be represented as:

Web/Mobile Clients → RESTful APIs → Cloud Backend → Cloud Storage

The infrastructure contains:

- **Frontend:** Web and mobile clients.
- **RESTful Web APIs:** Connect clients with backend services.
- **Cloud Platform:** Hosts backend application services.
- **Virtual Machines:** Provide computing resources.
- **Managed Kubernetes:** Supports scalability and reliability.
- **Cloud Storage:** Stores menu images, invoices, and customer data.
- **Security:** Authentication tokens, HTTPS, and role-based access control.

The case explicitly identifies RESTful APIs, cloud storage, virtual machines, managed Kubernetes, authentication tokens, HTTPS, and role-based access control.

3. Application of Cloud Concepts

When a customer places an order, the mobile or web client sends the request through a RESTful API to the backend service. The backend processes the order and stores the required information in cloud storage.

Managed Kubernetes allows the application to scale when the number of customers increases. For example, during peak meal times, additional resources can be used to handle increased traffic.

Conclusion:

The Online Food Ordering Platform effectively combines cloud computing, RESTful APIs, cloud storage, virtual machines, and managed Kubernetes. Authentication, HTTPS, and role-based access control provide security, while cloud scalability supports increasing customer demand.

Case Study Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution
Clarity	10%	Well-structured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized